Assignment. 2.

1.

```
Loop  li   x5, 20
      lw   x1, 1024(x5)
      add  x1, x1, x1
      sub  x2, x1, x3
      sw   x2, 1024(x5)
      subi x5, x5, 4
      bnez x5, loop
```

(a)

data ~~stall~~ hazard = 5 × 1 = 5

control hazard = 4 × 2 = 8

(b)

first time

```
li   x5, 20
lw   x1, 1024(x5)
add  x1, x1, x1
sub  x2, x1, x3
sw   x2, 1024(x5)
subi x5, x5, 4
benz x5, loop
```

second time

```
lw   x1, 1024(x5)
add  x1, x1, x1
sub  x2, x1, x3
sw   x2, 1024(x5)
subi x5, x5, 4
benz x5, loop
```

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		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	li	IF	ID	EX	MEM	WB														
2	lw		IF	ID	EX	MEM	WB													
3	add			IF	ID	ST	EX	MEM	WB											
4	sub				~~IF~~		IF	ID	EX	MEM	WB									
5	sw						IF	ID	EX	MEM										
6	subi							IF	ID	EX										
7	ben2								IF	ID	FL									
8	wrong path									IF	FL									
9	wrong path										IF	ID	EX	MEM	WB					
10	lw											IF	ID	ST ~~EX~~	EX ~~MEM~~	MEM ~~WB~~	WB			
11	add												IF	ST	ID	EX	MEM			
12	sub															ID	EX	MEM		
13	sw															IF	ID	EX	MEM	
14	subi																IF	ID	EX	MEM
15	ben2																	IF	ID	EX

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1c. li executes once.

loop instructions : 6

loop executes : 5

$N = 1 + 5 \times 6 = 31$

cycles = $N + 4 +$ stalls $= 31 + 4 + 5 + 8 = 48$

$CPI = \frac{48}{31} \approx 1.55$

2(a) If the processor supports delayed branch with 1 delay slot. we can get :

```
li   x5 20.
lw   x1 ,1024(x5)
loop : add  x1,x1,x1
       sub  x2, x1, x5
       sw   x2, 1024(x5)
       subi x5, x5, 4
       bnezd x5 loop
       lw   x1 1024(x5)
```

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0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
lw	IF	ID	EX	MEM	WB															
add		IF	ID	EX	MEM	WB														
sub			IF	ID	ST	EX	MEM	WB												
sw				IF	ST	ID	EX	MEM	WB											
bnez						IF	ID	EX	MEM	WB										
sub2							IF	ID	EX											
~~lw~~ strong								IF	ID	FL										
wrong ~~add~~									IF	FL										
lw										IF	ID	EX	MEM	WB						
add											IF	ID	ST	EX	MEM	WB				
sub												IF	ST	ID	~~EX~~ MEM	~~MEM~~ WB	WB			
sw														IF	ID	EX	MEM			
sub2															IF	ID	EX	MEM		
bnez																IF	ID	EX		

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2(a) -2.

$N = 1+1+5\times6 = 32$ Data stall $= 1$. control stall $= 4\times1 = 1$

cycles $= 32+4+1+4 = 41$

$CPI = \frac{41}{32} \approx 1.28$

2(b) $N = 31$ cycles $= 31+4+5+4 = 44$

$CPI = \frac{44}{31} \approx 1.42$

2(c) Original processor $C_{orig} = 48$

Delayed branch $C_{DB} = 41$

speedup $= \frac{48}{41} \approx 1.17$

BTB + 1-bit predictor

$C_{BTB} = 44$

speedup $= \frac{48}{44} \approx 1.09$